

Table 1. Laser-texturing used to functionalize the copper samples.

Sample	Pulse Length (ns)	Pulse Frequency (kHz)	Scanning Speed (mm s ⁻¹)	Average Fluence (J cm ⁻²)	Pattern
CH & CH-H	40	125	125	48.9	crosshatch (0° and 90°), spacing $\Delta x = 60 \mu\text{m}$
SC & SC-H	30	500	500	12.2	parallel lines (0°), cyclically variable spacing $\Delta x = \{35\text{--}65\} \mu\text{m}$
DC & DC-H	45	110	110	55.6	parallel lines (0°), cyclically variable spacing $\Delta x = \{35\text{--}65\} \mu\text{m}$

2.3. Surface Hydrophobization

Selected samples were hydrophobized using chemical vapor deposition (CVD) of a fluorinated silane. To perform the treatment, 0.05 mL of (heptadecafluoro-1,1,2,2-tetrahydrodecyl) trimethoxysilane (abbreviated to HTMS; Gelest Inc., Morrisville, PA, USA) was mixed with 0.95 mL of toluene ($\geq 99.7\%$, Honeywell International Inc., Charlotte, NC, USA) in a 5 mL glass vial at room temperature. The samples and the vial were placed in a plastic container,